NIAGARA 4 · API COMPARISON

Niagara N4 API Comparison

Built-in XML API · **Gline REST API v1 + v2** · Third-party connector · Built-in REST · Analytics

Side-by-side comparison of approaches to interact with Niagara N4 stations.

All Gline REST figures verified live on a Niagara 4.14 station.

FEATURE MATRIX

Feature comparison by channel

Feature	Built-in XML	Gline v1	Gline v2	Third-party connector	Built-in REST	Analytics
Read values	Yes	Yes	Yes	Yes	Yes	Read-only
Write values	Yes	Yes	Yes	Yes	Yes	No
Create / delete components	No	Yes	Yes	Yes	Limited	No
Deep copy (link rebuild)	No	Yes	Yes	No	No	No
Rename (link-safe)	No	Yes	Yes	No	No	No
Link / unlink	Partial	Yes	Yes	Yes	No	No
Invoke actions	Partial	Yes	Yes	Yes	No	No
Atomic batch (single POST)	No	11 ops	12 ops	Limited	No	No
Facet editing (units, enum ranges)	No	No	Yes	No	No	No
BQL query → JSON	No	No	Yes	No	No	Yes
Program management	No	No	Yes	No	No	No
Built-in auth (Basic)	Yes	No	Yes	Yes	Yes	Yes
CIDR allow-list	No	No	Yes	Partial	No	No
Alarm / history query	Yes (best)	No	No	Partial	Partial	Yes
License cost	Free	Free	On request	Per station	Free	Subscription
SMA required	No	No	No	No	Yes	No
Port	—	8081	8082	—	—	—

** v2 is distributed on request. Both Gline editions are independent modules and can run side by side.*

GLINE REST API V2 NEW in v2

What v2 adds

v2 = v1 + authentication + access control + BQL + program management + facet editing.

BQL query → JSON

Run BQL over HTTP and get a structured table: `{"columns":[…],"rows":[…],"count":N}`. Ready for reports, dashboards, and AI pipelines — no XML parsing.

Program management

`POST /api/program/write`, `POST /api/program/addslot`, `GET /api/program/status` — write source, add slots, read status over HTTP.

Facet editing (v2.4)

Write facets on any point or property — numeric (`min/max/precision`), `units`, and enum `range=Low,Mid,High`. Read a specific slot's facets back.

Built-in authentication

HTTP Basic Auth, configurable at runtime. Missing or wrong credentials → **401**.

CIDR allow-list

Restrict access to trusted network segments (comma-separated ranges). Outside the list → **403**. Loopback always allowed as a recovery path.

Persistent config

Auth and allow-list configuration survives station restarts. No external reverse proxy needed.

Verified live on Niagara 4.14. 40 / 40 test cases passed (29 functional + 11 access-control).

ENDPOINTS

HTTP endpoints

Endpoint	Method	Edition	Purpose
`/api/status`	GET	**v1 · v2**	Service status (auth-exempt liveness probe)
`/api/modules`	GET	**v1 · v2**	Module information
`/api/types`	GET	**v1 · v2**	Component types
`/api/config`	GET	**v1 · v2**	Browse the station tree
`/api/batch`	POST	**v1 · v2**	Batch component operations + BQL
`/api/program/write`	POST	**v2**	Write program source
`/api/program/addslot`	POST	**v2**	Add a program slot
`/api/program/status`	GET	**v2**	Read program status

- **Batch operations · v1:** ADD · LINK · RENAME · DUPLICATE · CP · MOD · UNLINK · REMOVE · READ · LIST · INVOKE
- **Batch operations · v2:** the 11 above + BQL (and facet editing)

Example — BQL to JSON (v2):

```
curl -u "<USER>:<PASS>" -X POST http://<HOST>:8082/api/batch \
  -d '[{"OP":"bql","Q":"select * from control:NumericWritable"}]'
```

DECISION GUIDE

Which to use

Gline REST API v2

- Component lifecycle **and** auth/access control in one module
- BQL results as ready-to-use JSON tables
- Facet editing (units, enum ranges, precision)
- Program management over HTTP
- 11 batch operations + BQL in one atomic request

Gline REST API v1

- 11 batch operations, no authentication layer
- Single `.jar` drop-in, no SMA required
- Best for trusted networks and quick automation

Built-in XML API

- Built-in, free alarm and history extraction
- Best-in-class for alarms and trends

Built-in REST / third-party tools

- **Built-in REST:** simple JSON read/write where the station has an SMA
- **Third-party connectors:** commercial licensing and support
- **Analytics:** dashboards and reporting, not station management

Use the built-in XML API for alarms and history; use Gline REST API for everything else.

CONTACT

Let's build your automation

Commercial customization, integration, or deployment support.

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All Gline REST figures verified live on a Niagara 4.14 station. Features, availability and pricing of third-party products may vary — verify directly with each vendor.